



OOAD Mid Term-answers - Mid paper

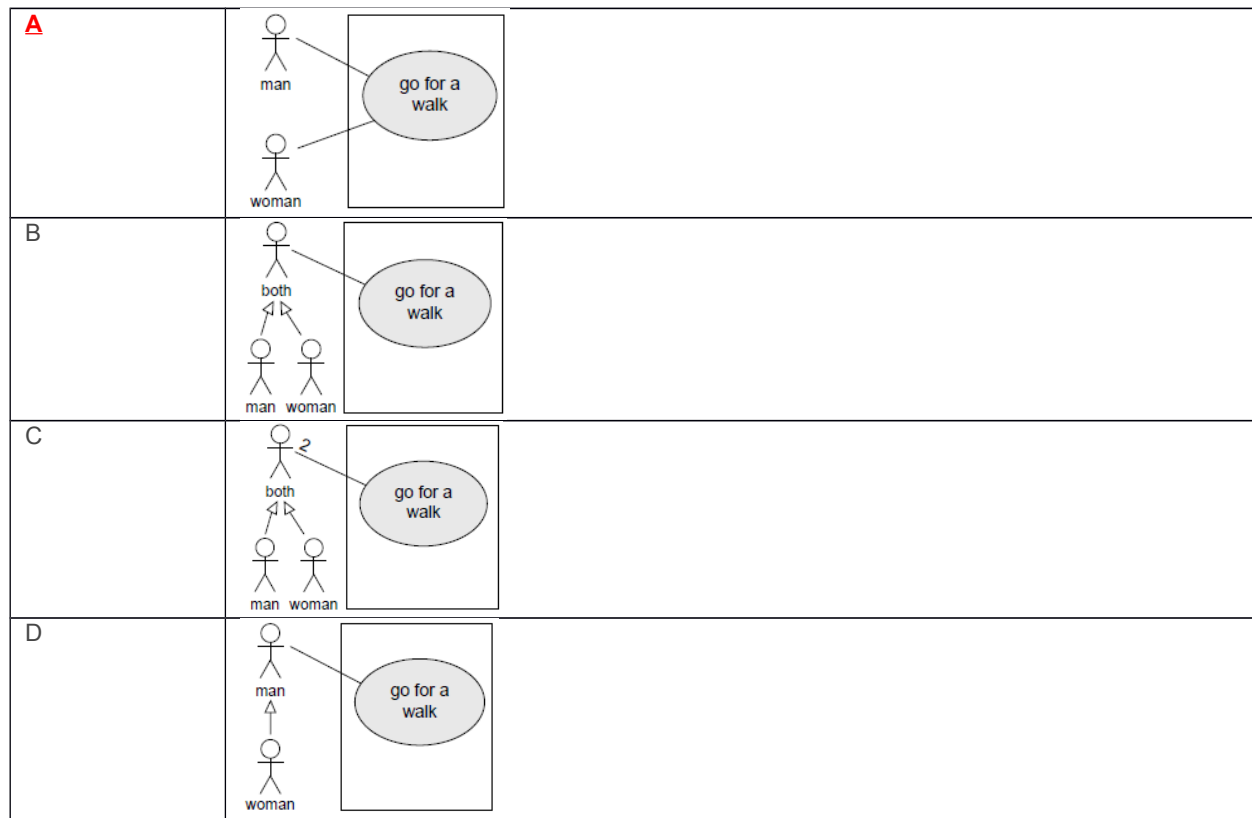
Object-Oriented Design (George Washington University)



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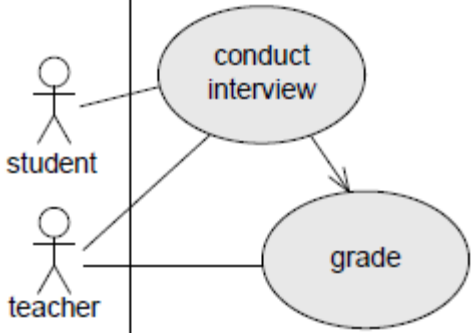
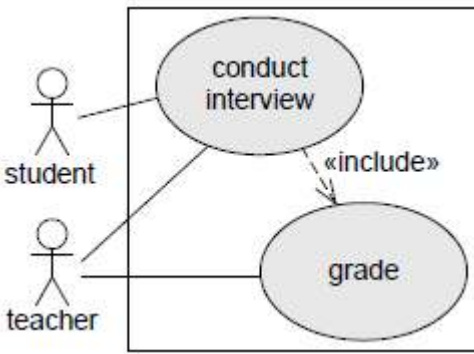
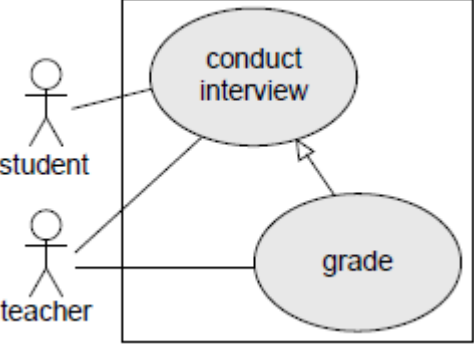
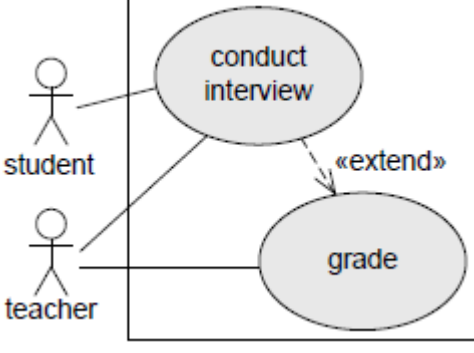
Q1) How do you model the following situation with a UML2 use case diagram:

A man and a woman go for a walk.



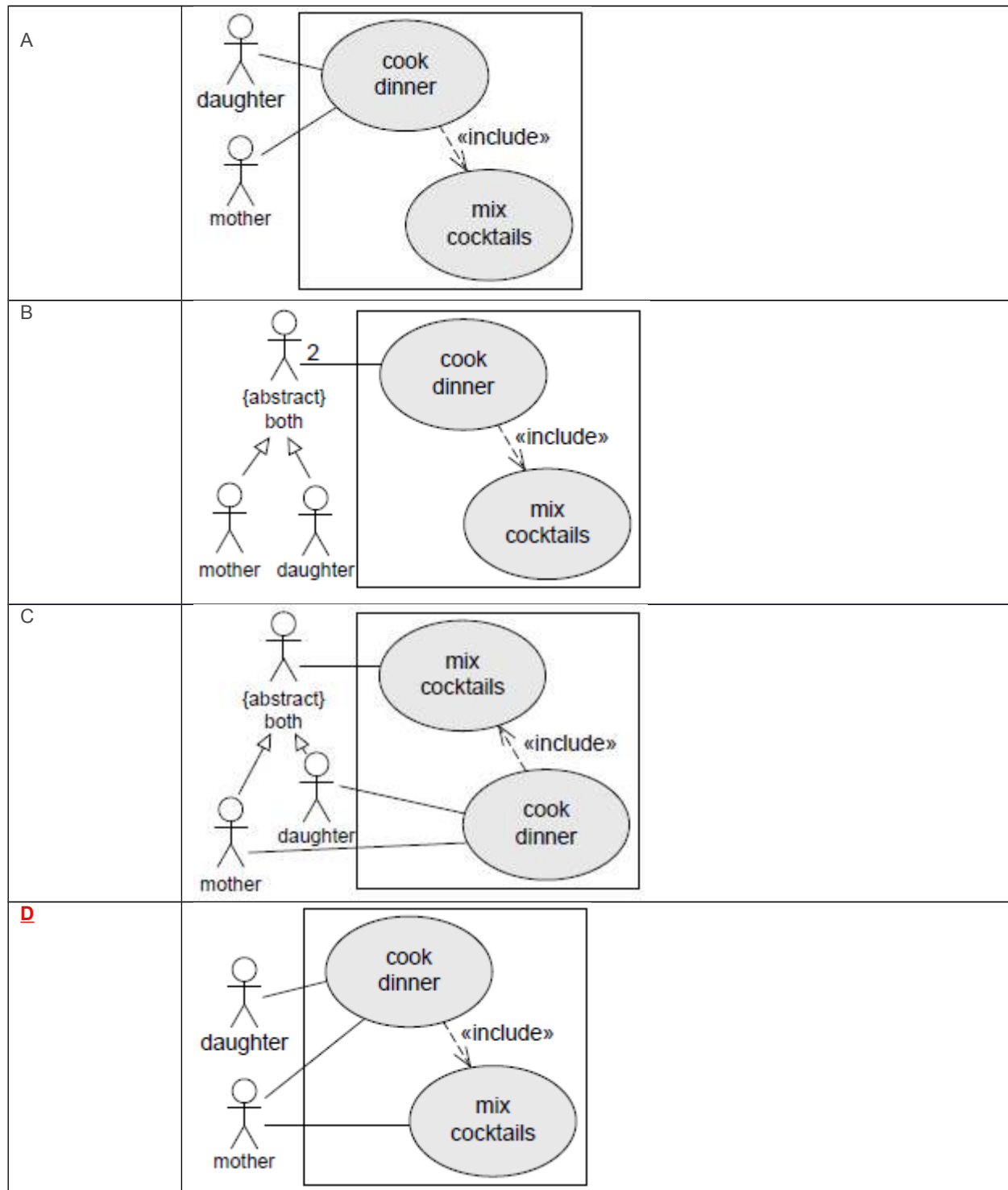
Q2) How do you model the following situation with a UML2 use case diagram:

A teacher is conducting an interview with a student. In the course of that, the teacher always has to grade the student.

A		
B		
C		
D		

Q3) How do you model the following situation with a UML2 use case diagram:

A mother cooks dinner together with her daughter. In the course of that, the mother also always has to mix the cocktails.



Q4) How do you model the following situation with a UML2 use case diagram:

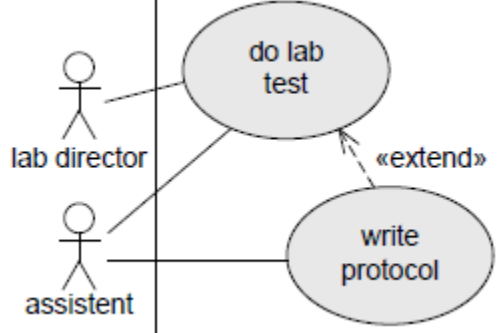
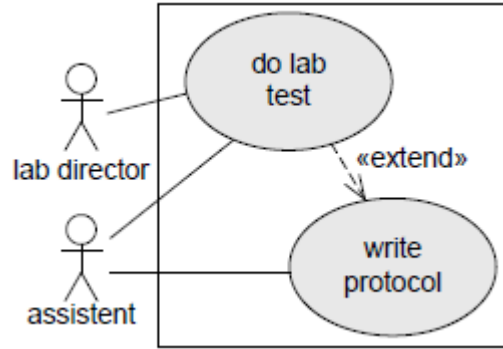
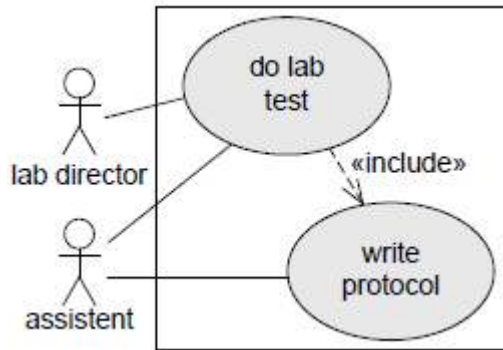
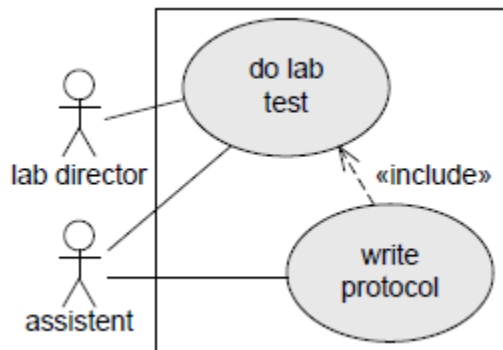
The lab director does a lab test together with his assistant. The assistant always has to write a protocol during the lab test.

A	<pre>graph TD LD((lab director)) --- UC1((do lab test)) AS((assistant)) --- UC1 UC1 -.-> «extend» UC2((write protocol))</pre>
B	<pre>graph TD LD((lab director)) --- UC1((do lab test)) AS((assistant)) --- UC1 UC1 -.-> «extend» UC2((write protocol))</pre>
C	<pre>graph TD LD((lab director)) --- UC1((do lab test)) AS((assistant)) --- UC1 UC1 -.-> «include» UC2((write protocol))</pre>
D	<pre>graph TD LD((lab director)) --- UC1((do lab test)) AS((assistant)) --- UC1 UC2((write protocol)) -.-> «include» UC1</pre>

Q5) How do you model the following situation with a UML2 use case diagram:

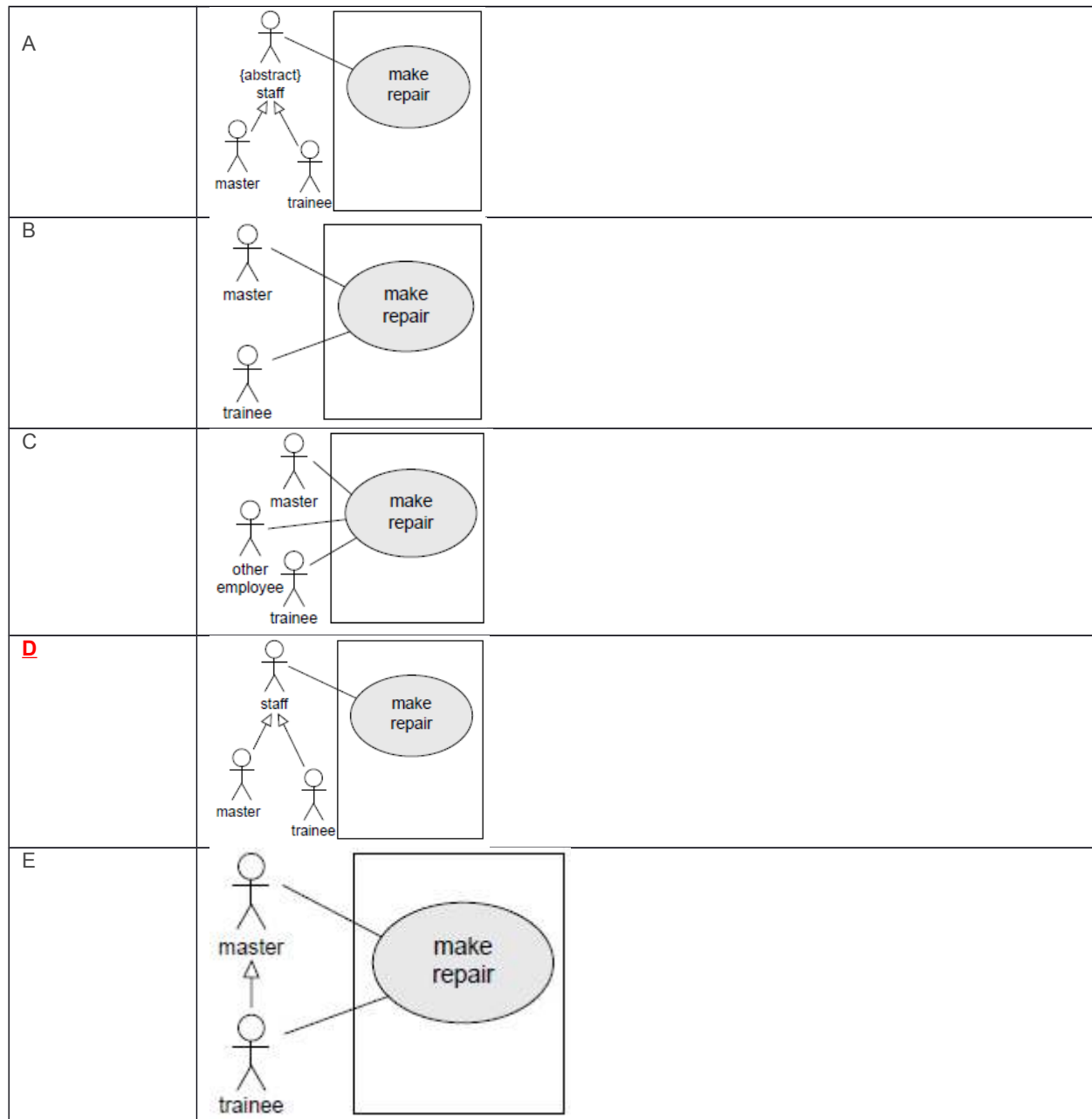
The lab director does a lab test together with his assistant. The assistant always has to write a protocol during the lab test.

The lab director does a lab test together with his assistant. The assistant always has to write a protocol during the lab test.

A		
B		
C		
D		

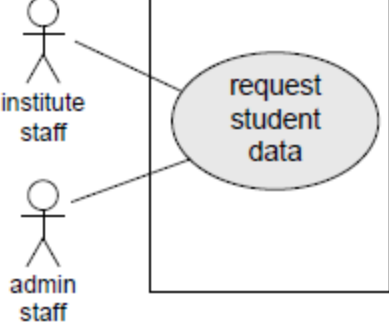
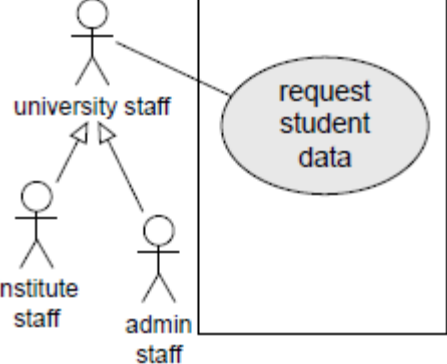
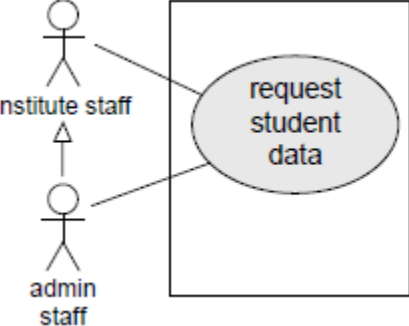
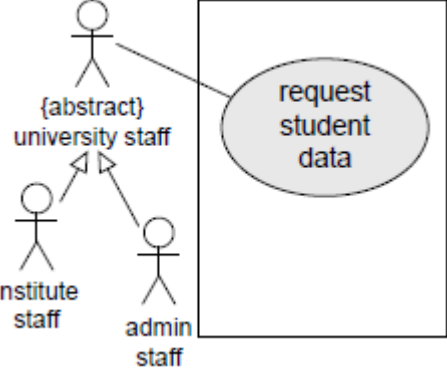
Q6) How do you model the following situation with a UML2 use case diagram:

A repair can be made by a master, a trainee or any other repair shop employee.



Q7) How do you model the following situation with a UML2 use case diagram:

Student data is requested by either institute staff or administration staff.

A	 <pre> graph LR Institute[institute staff] --- UC((request student data)) Admin[admin staff] --- UC </pre>
B	 <pre> graph LR Institute[institute staff] -- > US[university staff] Admin[admin staff] -- > US US --- UC((request student data)) </pre>
C	 <pre> graph LR Admin[admin staff] -- > Institute[institute staff] Institute --- UC((request student data)) Admin --- UC </pre>
D	 <pre> graph LR Institute[institute staff] -- > US["{abstract} university staff"] Admin[admin staff] -- > US US --- UC((request student data)) </pre>
E	All answer are wrong

Q8) How do you model the following situation with a UML2 use case diagram:

Mother and daughter cook dinner together. Sometimes it is necessary that one of them gets chives from the garden.

A	<pre> graph LR mother --> get_chives mother --> cook daughter --> cook get_chives -.-> «extend» cook </pre>
B	<pre> graph LR mother --> get_chives daughter --> cook get_chives -.-> «extend» cook </pre>
C	<pre> graph LR mother --> get_chives daughter --> cook get_chives -.-> «extend» cook </pre>
D	<pre> graph LR mother --> get_chives daughter --> cook get_chives -.-> «extend» cook </pre>
E	All answer are wrong

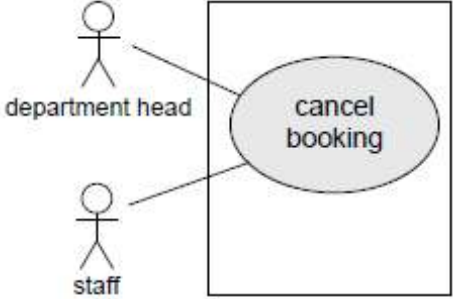
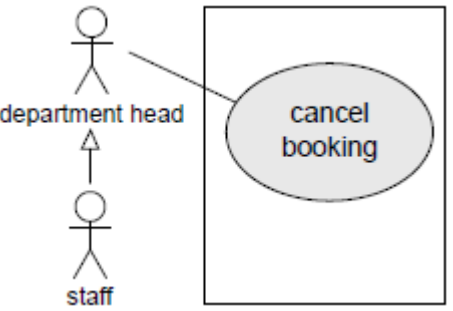
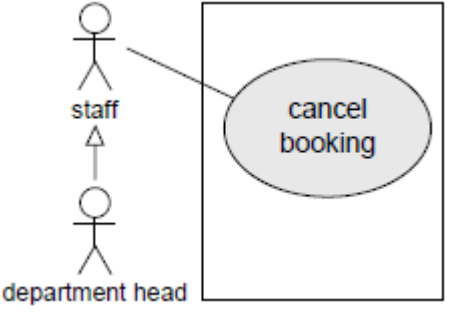
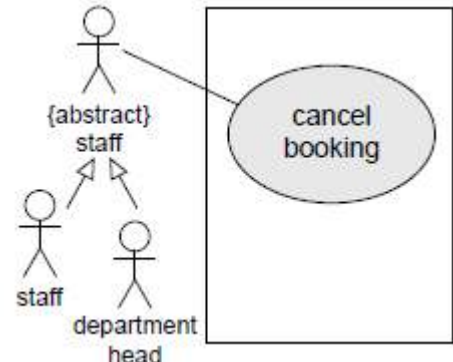
Q9) How do you model the following situation with a UML2 use case diagram:

A person goes for lunch. In the course of that it might be necessary that the person withdraws cash from an ATM.

A	<pre> graph TD person((person)) --- go_for_lunch((go for lunch)) person --- withdraw_cash((withdraw cash)) go_for_lunch -.-> «include» withdraw_cash </pre>
B	<pre> graph TD person((person)) --- go_for_lunch((go for lunch)) person --- withdraw_cash((withdraw cash)) withdraw_cash -.-> «include» go_for_lunch </pre>
C	<pre> graph TD person((person)) --- go_for_lunch((go for lunch)) person --- withdraw_cash((withdraw cash)) withdraw_cash --> go_for_lunch </pre>
D	<pre> graph TD person((person)) --- go_for_lunch((go for lunch)) person --- withdraw_cash((withdraw cash)) go_for_lunch -.-> «extend» withdraw_cash </pre>
E	<pre> graph TD person((person)) --- go_for_lunch((go for lunch)) person --- withdraw_cash((withdraw cash)) go_for_lunch --> withdraw_cash </pre>
F	All answer are wrong

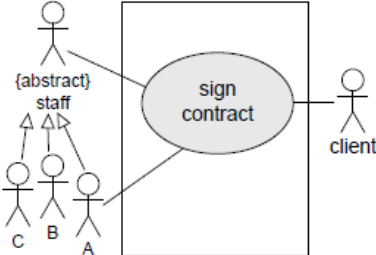
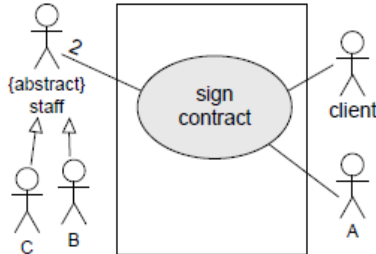
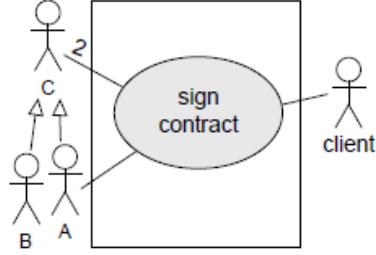
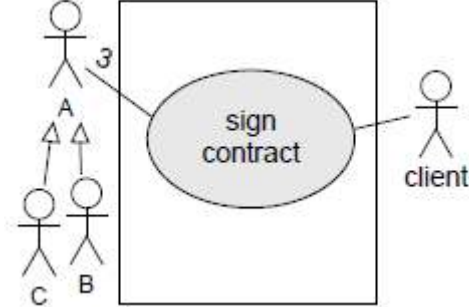
Q10) How do you model the following situation with a UML2 use case diagram:

A person goes for lunch. In the course of that it might be necessary that the person withdraws cash from an ATM.

A	
B	
C	
D	
E	A, b, c, d are wrong
F	A, b, c, d are correct

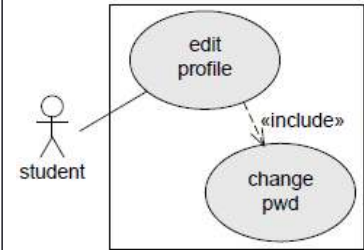
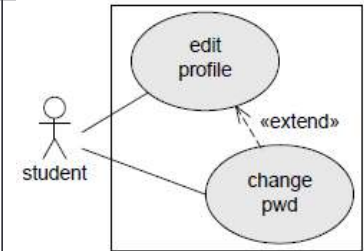
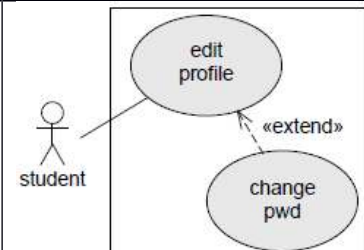
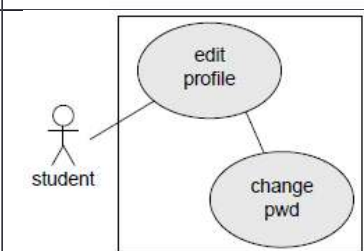
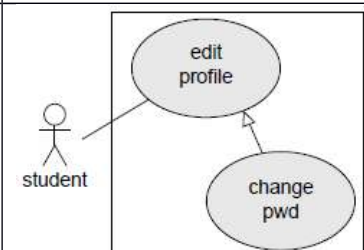
Q11) How do you model the following situation with a UML2 use case diagram:

There are 3 different types of staff members namely A, B and C. For a valid contract, the client, one staff member of Type A and two other staff members of either type (A, B or C) have to sign the contract.
(Possible employee combinations: AAA, AAB, AAC, ABB, ACC, ABC)

A	
B	
C	
D	
E	A, b, c, d are wrong
F	A, b, c, d are correct

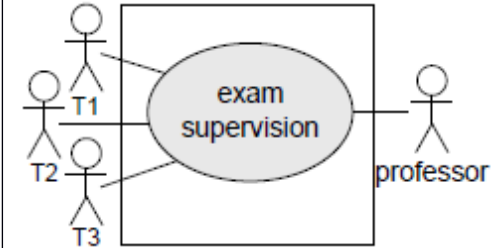
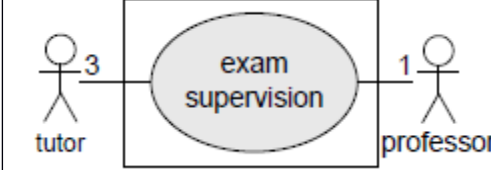
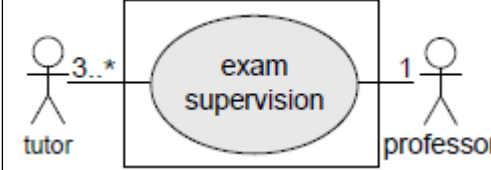
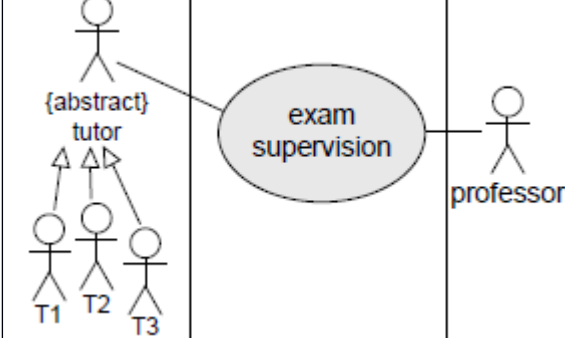
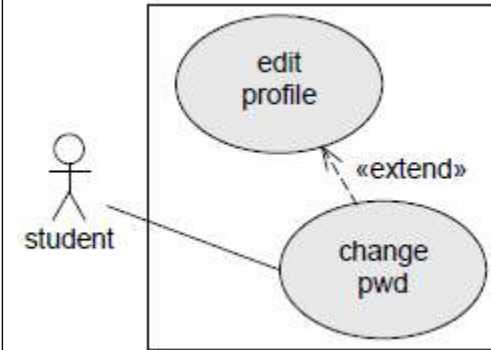
Q12) How do you model the following situation with a UML2 use case diagram:

A student edits his user profile. In the course of that he can also change his password if he likes.

A		In this approach, change pwd has to be executed every time edit profile is executed.
B		
C		
D		The line between edit profile and change pwd does not comply with the UML2 standard, there should be modeled an extend relationship instead.
E		
F	A, b, c, d, and e are wrong	

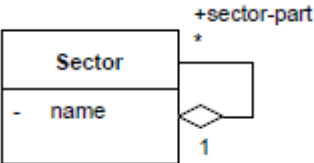
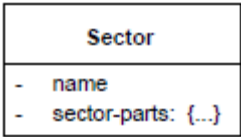
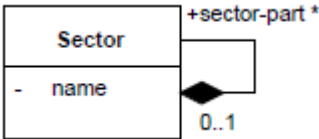
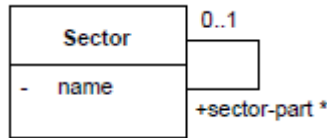
Q13) How do you model the following situation with a UML2 use case diagram:

An exam supervision is done by one professor and three tutors.

A	
B	
C	
D	
E	
F	A, b, c, d, and e are wrong

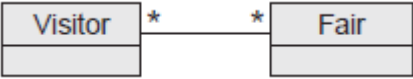
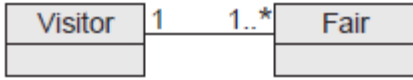
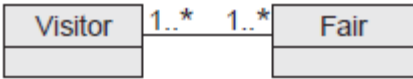
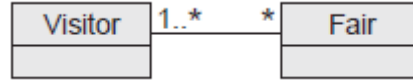
Q14) How do you model the following situation with a UML2 class diagram:

A fitness center consists of several sectors. One sector may be divided into several sectors which might as well be divided into sectors and so on.

A	
B	
C	
D	
E	A, b, c, d are wrong
F	A, b, c, d are correct

Q15) How do you model the following situation with a UML2 class diagram:

A fair is visited by at least one visitor. One visitor visits at least one fair.

A	
B	
C	
D	
E	A, b, c, d are wrong
F	A, b, c, d are correct

Q16) How do you model the following situation with a UML2 class diagram:

Every product is in one of the three following states: solid, liquid or gas.

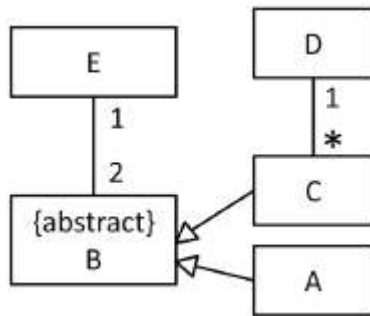
A	<pre> classDiagram class Product class Solid class Liquid class Gas Product "*" -- "*" Solid Product "*" -- "1" Liquid Product "*" -- "7" Gas Product ..> Solid : {xor} Product ..> Liquid : {xor} Product ..> Gas : {xor} </pre>
B	<pre> classDiagram class Product { solid: boolean liquid: boolean gas: boolean } </pre>
C	<pre> classDiagram class Product { state: String } </pre>
D	<pre> classDiagram class Product { s: state } class State { <<enumeration>> solid liquid gas } </pre>
E	A, b, c, d are wrong
F	A, b, c, d are correct

Q17) How do you model the following situation with a UML2 class diagram:

Each talk is done by a professor or by an assistant.

A	<pre> classDiagram class Talk class Professor class Assistant Talk "*" -- "1" Professor Talk "*" -- "1" Assistant </pre>
B	<pre> classDiagram class Talk class Professor class Assistant Talk "*" -- "1" Professor Talk "*" -- "1" Assistant Note over Talk, Professor, Assistant: {xor} </pre>
C	<pre> classDiagram class Product { state: String } </pre>
D	<pre> classDiagram class Talk class Staff { <<abstract>> } class Professor class Assistant Talk "*" -- "1" Staff Professor -- > Staff Assistant -- > Staff </pre>
E	<pre> classDiagram class Talk { lecturer: String } </pre>
F	A, b, c,d, and e are wrong

Q18) You are given the following clipping of a UML2 class diagram. Which of the following object diagrams are consistent with the class diagram?



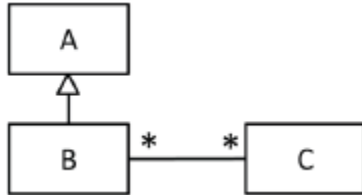
A	
B	
C	
D	
E	
F	A, b, c, d, and e are wrong

Q19) How do you model the following situation with a UML2 class diagram:

Furniture Fritz charges custom-made furniture which is assembled from prefabricated components.

A	<pre> classDiagram class Bill class Custom-made-furniture class Component Bill "1" -- "*" Custom-made-furniture Component < -- Custom-made-furniture </pre>
B	<pre> classDiagram class Bill class Custom-made-furniture class Component Bill "1" -- "*" Custom-made-furniture Custom-made-furniture "1" *-- "*" Component </pre>
C	<pre> classDiagram class Bill class Custom-made-furniture class Component Bill "*" -- "1" Custom-made-furniture Custom-made-furniture "1" *-- "*" Component </pre>
D	<pre> classDiagram class Bill class Custom-made-furniture class Component Bill "1" -- "*" Custom-made-furniture Custom-made-furniture "1" *-- "*" Component </pre>
E	A, b, c, d are wrong
F	A, b, c, d are correct

Q20) You are given the following clipping of a UML2 class diagram. Which of the following object diagrams are consistent with the class diagram?



A	
B	
C	
D	
E	
F	A, b, c, d, and e are wrong

Q21) How do you model the following situation with a UML2 class diagram:

There are multiple different bird species, e.g. blackbird, thrush and starling.

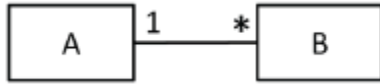
A	<pre> classDiagram class Bird { t: typ } class EnumType["«enumeration» typ"] { blackbird thrush starling } Bird --> EnumType </pre>
B	<pre> classDiagram class Bird { <<abstract>> } class Starling class Blackbird class Thrush Bird "1" -- "*" Starling Bird "1" -- "*" Blackbird Bird "1" -- "*" Thrush </pre>
C	<pre> classDiagram class Bird class Starling class Blackbird class Thrush Bird < -- Starling Bird < -- Blackbird Bird < -- Thrush </pre>
D	<pre> classDiagram class Bird { <<abstract>> } class Starling class Blackbird class Thrush Bird < -- Starling Bird < -- Blackbird Bird < -- Thrush </pre>
E	A, b, c, d are wrong
F	A, b, c, d are correct

Q22) How do you model the following situation with a UML2 class diagram:

There are exactly two kinds of participants, namely members and guests. Guests are invited by members.

A	<pre> classDiagram class Participant["{abstract} Participant"] class Member class Guest Participant < -- Member Participant < -- Guest Member "1" -- "*" Guest : invite </pre>
B	<pre> classDiagram class invite class Participant class Member class Guest invite < -- Member invite < -- Guest invite "*" -- "1" Participant Member "1" -- "*" Guest : invite </pre>
C	<pre> classDiagram class Participant["{abstract} Participant"] class Member class Guest class invite Participant < -- Member Participant < -- Guest Member "1" -- "1" Guest : invite </pre>
D	<pre> classDiagram class Participant class Member class Guest Participant < -- Member Participant < -- Guest Member "1" -- "*" Guest : invite </pre>
E	A, b, c, d are wrong
F	A, b, c, d are correct

Q23) You are given the following clipping of a UML2 class diagram. Which of the following object diagrams are consistent with the class diagram?



A	<pre> graph LR a1["a1 : A"] --- b2["b2 : B"] a2["a2 : A"] --- b1["b1 : B"] </pre>
B	<pre> graph LR a1["a1 : A"] --- b1["b1 : B"] a1 --- b2["b2 : B"] </pre>
C	<pre> graph LR b1["b1 : B"] </pre>
D	<pre> graph LR a1["a1 : A"] </pre>
E	<pre> graph LR a1["a1 : A"] --- b1["b1 : B"] a2["a2 : A"] --- b1 </pre>
F	<pre> graph LR a1["a1 : A"] --- b1["b1 : B"] </pre>

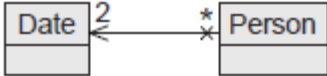
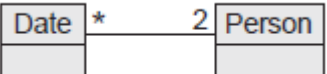
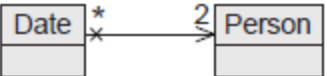
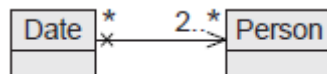
Q24) How do you model the following situation with a UML2 class diagram:

During one soccer season, multiple players participate in multiple games. Each player scores in each game a certain number of goals.

A	
B	
C	
D	
E	A, b, c, d are wrong
F	A, b, c, d are correct

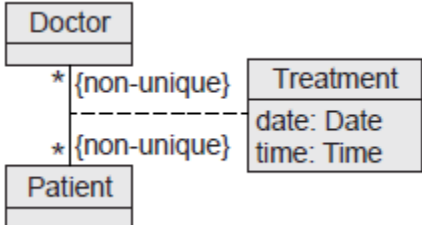
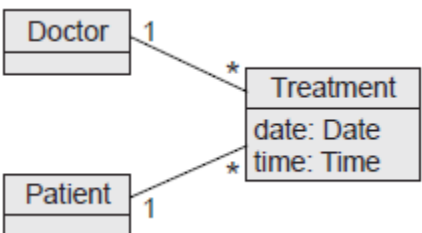
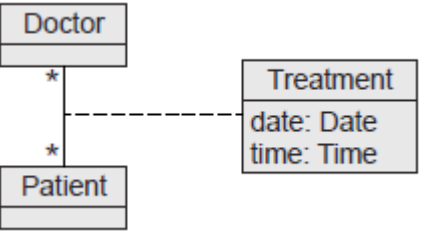
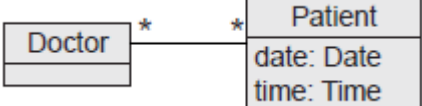
Q25) How do you model the following situation with a UML2 class diagram:

On a blind date exactly two people take part. A person can take part in multiple blind dates. It is not sufficient to know a person to derive all his/her dates.

A	
B	
C	
D	
E	A, b, c, d are wrong
F	A, b, c, d are correct

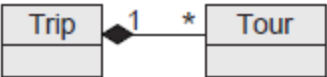
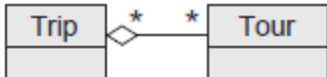
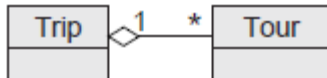
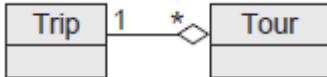
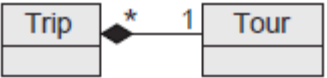
Q26) How do you model the following situation with a UML2 class diagram:

A doctor treats multiple patients, a patient can be treated by multiple doctors. To document which doctor has when treated which patient, the date and time of each treatment are stored. It has to be possible that a doctor treats the same patient multiple times.

A	 <pre> classDiagram class Doctor class Patient class Treatment { date: Date time: Time } Doctor "*" -- "*" Treatment : {non-unique} Patient "*" -- "*" Treatment : {non-unique} </pre>
B	 <pre> classDiagram class Doctor class Patient class Treatment { date: Date time: Time } Doctor "1" -- "*" Treatment Patient "1" -- "*" Treatment </pre>
C	 <pre> classDiagram class Doctor class Patient class Treatment { date: Date time: Time } Doctor "*" -- "*" Treatment Patient "*" -- "*" Treatment </pre>
D	 <pre> classDiagram class Doctor class Patient class Treatment { date: Date time: Time } Doctor "*" -- "*" Treatment Patient "*" -- "*" Treatment </pre>
E	A, b, c, d are wrong
F	A, b, c, d are correct

Q27) How do you model the following situation with a UML2 class diagram:

A trip might comprise of multiple tours, one tour can be included in several trips.

A	
B	
C	
D	
E	
F	

Q28) How do you model the following situation with a UML2 class diagram:

An event may be part of another event.

A	<pre> classDiagram class Event { +eventpart * } Event "*" -- "0..1" Event : +eventpart </pre>
B	<pre> classDiagram class Event { +eventpart * } Event "*" -- "0..1" Event : +eventpart </pre>
C	<pre> classDiagram class Event { +eventpart * } Event "*" -- "0..1" Event : +eventpart </pre>
D	<pre> classDiagram class Event { - eventparts: {...} } </pre>
E	A, b, c, d are wrong
F	A, b, c, d are correct

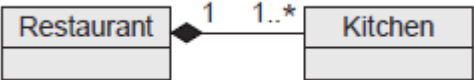
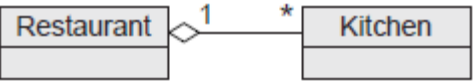
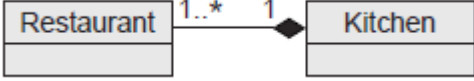
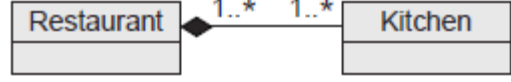
Q29) How do you model the following situation with a UML2 class diagram:

A vehicle comprises of multiple parts. Each part can be built in a maximum of one super-part. One super-part comprises of multiple sub-parts

A	<pre> classDiagram class Part class Super class Sub class Vehicle Part < -- Super Part < -- Sub Part "1" *-- "*" Vehicle </pre>
B	<pre> classDiagram class Part class Super class Sub class Vehicle Part < -- Super Part < -- Sub Part "1" *-- "*" Vehicle Part "0..1" *-- "*" Part : part of </pre>
C	<pre> classDiagram class Part class Super class Sub class Vehicle Part < -- Super Part < -- Sub Part "1" *-- "*" Vehicle Part "0..1" *-- "*" Part : part of </pre>
D	<pre> classDiagram class Part class Super class Sub class Vehicle Part < -- Super Part < -- Sub Part "1" *-- "*" Vehicle Part "*" *-- "0..1" Part : part of </pre>
E	A, b, c, d are wrong
F	A, b, c, d are correct

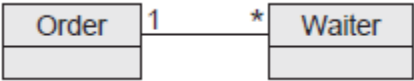
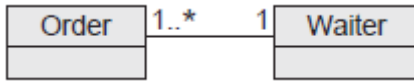
Q30) How do you model the following situation with a UML2 class diagram:

Every restaurant has at least one kitchen, one kitchen is part of exactly one restaurant.

A	
B	
C	
	
E	A, b, c, d are wrong
F	A, b, c, d are correct

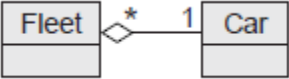
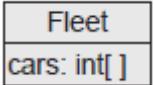
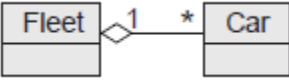
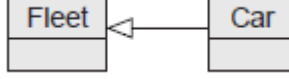
Q31) How do you model the following situation with a UML2 class diagram:

An order is made with exactly one waiter, one waiter handles multiple orders.

A	
B	
C	
D	
E	A, b, c, d are wrong
F	A, b, c, d are correct

Q32) How do you model the following situation with a UML2 class diagram:

The car fleet of a car rental contains multiple cars, one car belongs to exactly one car fleet.

A	
B	
C	
D	
E	A, b, c, d are wrong
F	A, b, c, d are correct

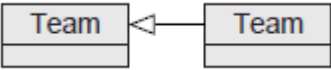
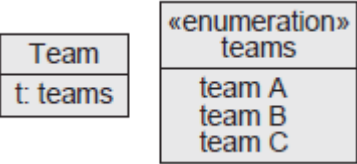
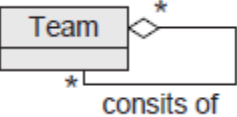
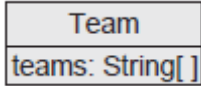
Q33) How do you model the following situation with a UML2 class diagram:

A season worker may be employed in one or several seasons, the beginning and the end of each employment period is saved.

A	<pre> classDiagram class SeasonalWorker class Season { - beginning - end } SeasonalWorker "*" -- "1..*" Season </pre>
B	<pre> classDiagram class SeasonWorker { - beginning - end } </pre>
C	<pre> classDiagram class Employment { - beginning - end } class SeasonalWorker class Season Employment .. SeasonalWorker Employment .. Season SeasonalWorker "*" -- "1..*" Season </pre>
D	<pre> classDiagram class Employment { - beginning - end } class SeasonalWorker class Season Employment "1" *-- "1..*" SeasonalWorker Employment "1" *-- "1..*" Season </pre>
E	A, b, c, d are wrong
F	A, b, c, d are correct

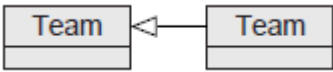
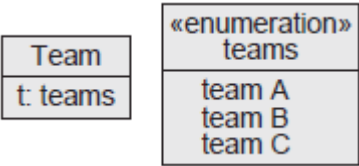
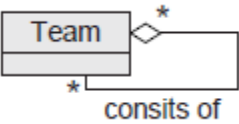
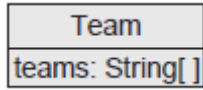
Q34) How do you model the following situation with a UML2 class diagram:

One team may consist of several other teams.

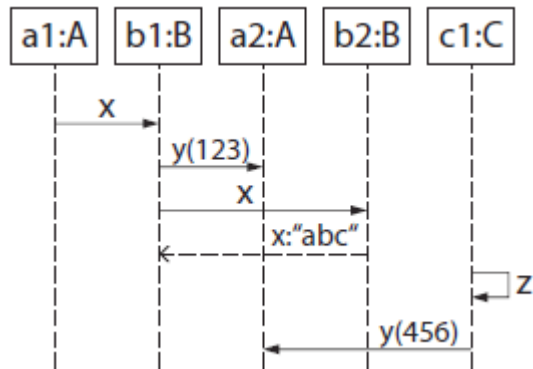
A	
B	
C	
D	
E	A, b, c, d are wrong
F	A, b, c, d are correct

Q35) How do you model the following situation with a UML2 class diagram:

Teams may consist of several other teams.

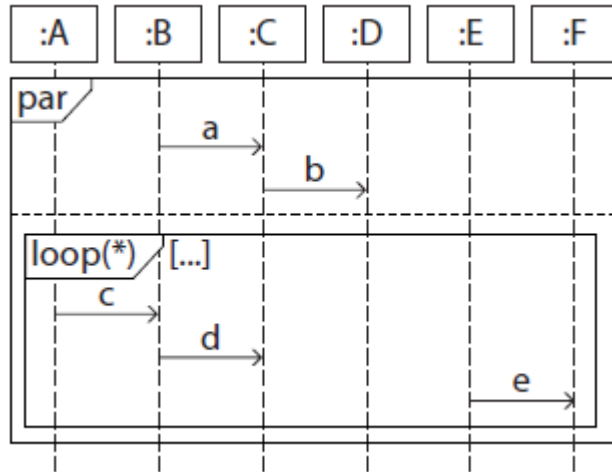
A	
B	
<u>C</u>	
D	
E	A, b, c, d are wrong
F	A, b, c, d are correct

Q36) You are given the following sequence diagram. Which operations does class C have according to the diagram?



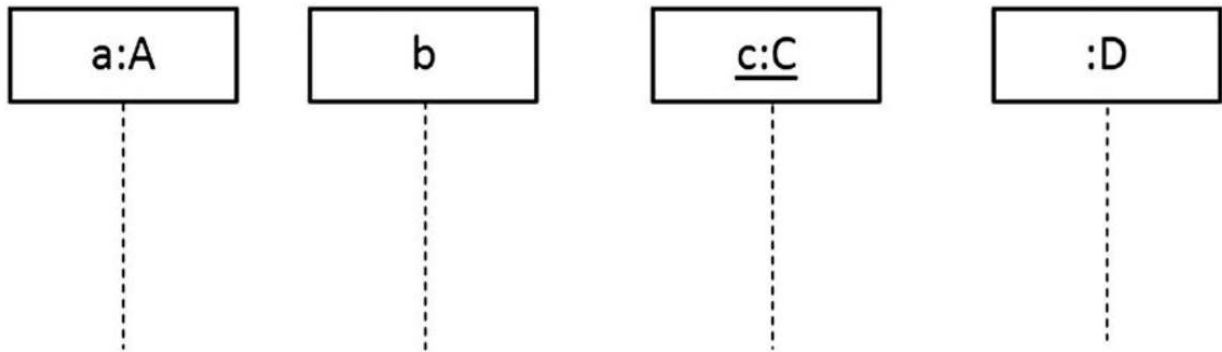
A	<code>x():String</code>
B	<code>z():int</code>
C	<code>x():int</code>
D	<code>x(String):void</code>
E	<code>y():int</code>
F	<code>x(void)</code>
G	<code>y(int):void</code>
H	<code>z():void</code>
I	<code>y():void</code>
J	<code>x():void</code>

Q37) You are given the following sequence diagram. Which traces are possible? (Give a point to this question to all students)



A	$b \rightarrow a$
B	$a \rightarrow c \rightarrow d \rightarrow e \rightarrow b$
C	$a \rightarrow c \rightarrow d \rightarrow b \rightarrow e$
D	$a \rightarrow b \rightarrow c \rightarrow d \rightarrow e \rightarrow a \rightarrow b$
E	$c \rightarrow d \rightarrow e \rightarrow a \rightarrow b$
F	$c \rightarrow a \rightarrow b \rightarrow d \rightarrow e$

Q38) Given the following sequence diagram, which one of the following potential lifelines needs to be changed so this diagram can be corrected?



A. A

B. B

C. C

D. D

Discussion:

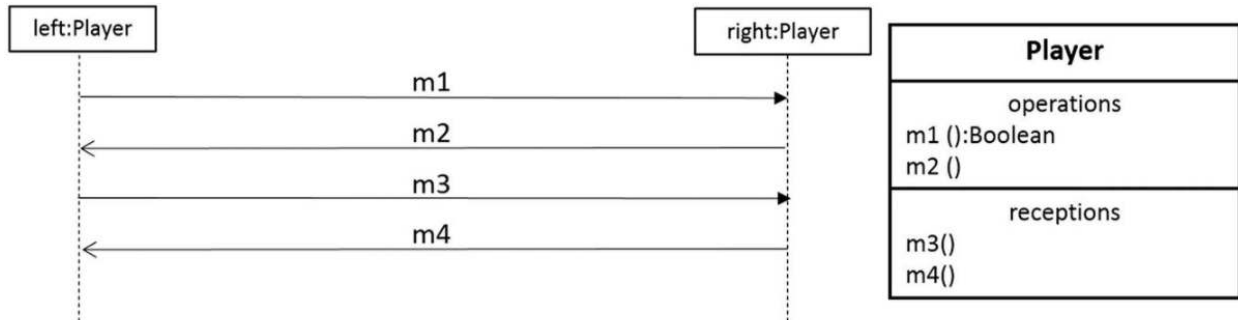
A—No, this is the normal lifeline.

B—No, the type or class name, the part of the name that follows the role name and is preceded by the ":" is an optional field.

C—Yes, instances are not connectable elements. You can see that it is an instance by the underline.

D—No, the role name, the part of the name that precedes the ":", is not a required field.

Q39) Which of the messages in the below diagram is not compatible with the definitions shown in the class **Player**?



- A. m1
- B. m2
- C. m3
- D. m4

Messages m2 and m4 are asynchronous messages because of their v-shaped arrowhead.

They may be signals (shown in the receptions compartment) or asynchronous operation calls.

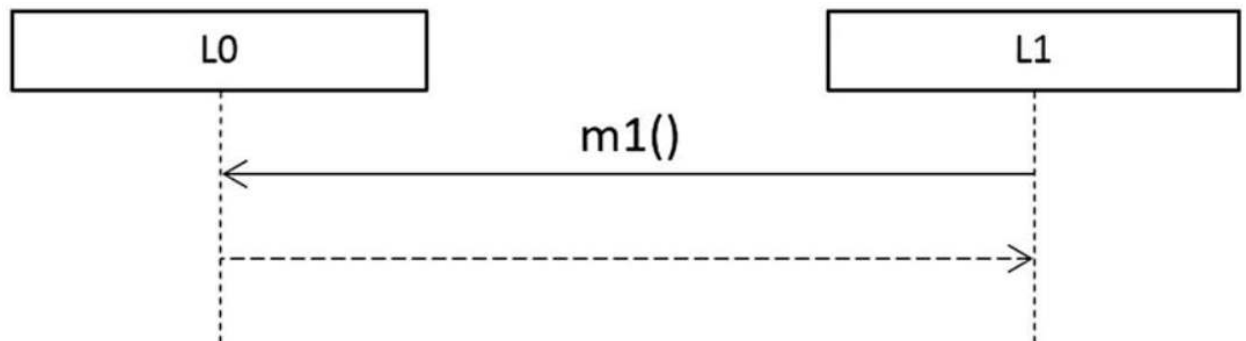
A—No, m1 is compatible with the m1 operation on the Player. It must be a synchronous call because it has a return type.

B—No, m2 is an asynchronous message because it has a v-shaped arrowhead. In this case, it has a compatible match in the operations compartment. This is an example of an asynchronous operations call.

C—Yes, m3 is not compatible. The only match in Player is a reception. However, a reception must be an asynchronous signal. However, the arrowhead indicates that it is a synchronous message.

D—No, m4 is compatible with the reception in Player.

Q40) What needs to be changed in the following Sequence Diagram? (Eliminate question!!!)



- A. The empty () needs to be removed or filled in.
- B. The first message should move from right to left.
- C. The arrowhead on the dashed line should be made into a solid arrowhead.
- D. The m1 is a signal and cannot have a return. Instead, change m1 into a synchronous operation call by making it have a solid arrowhead.

Discussion:

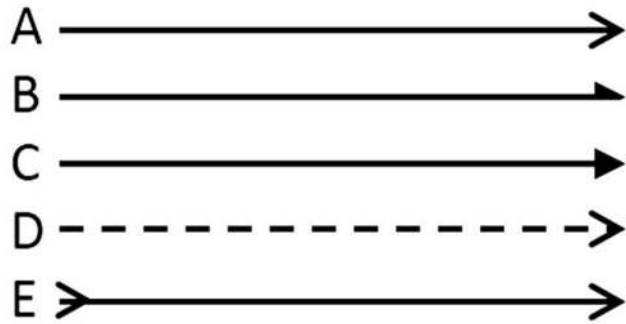
A—No, While messages usually have () after the name this is not needed and the () need not be filled in.

B—No, the direction is acceptable.

C—No, that doesn't accomplish anything.

D—Yes, a signal (asynchronous) can't have a return, so if you change the signal to an operation (synchronous) it would be correct.

Q41) Which arrow format below indicates an asynchronous message?



- A. A
- B. B
- C. C
- D. D
- E. E

Discussion:

A—Yes, this is correct.

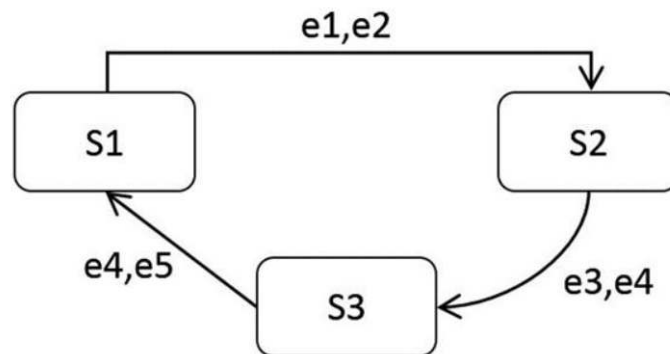
B—No, this was the format for asynchronous messages in UML 1.x.

C—No, this is the format for synchronous messages.

D—No, this is the format for return messages.

E—No, UML does not have tails on its arrows.

Q42) Examine the State Machine Diagram below. What state will the State Machine be in after the events e1, e2, e3, and e1 occur (in that order)?



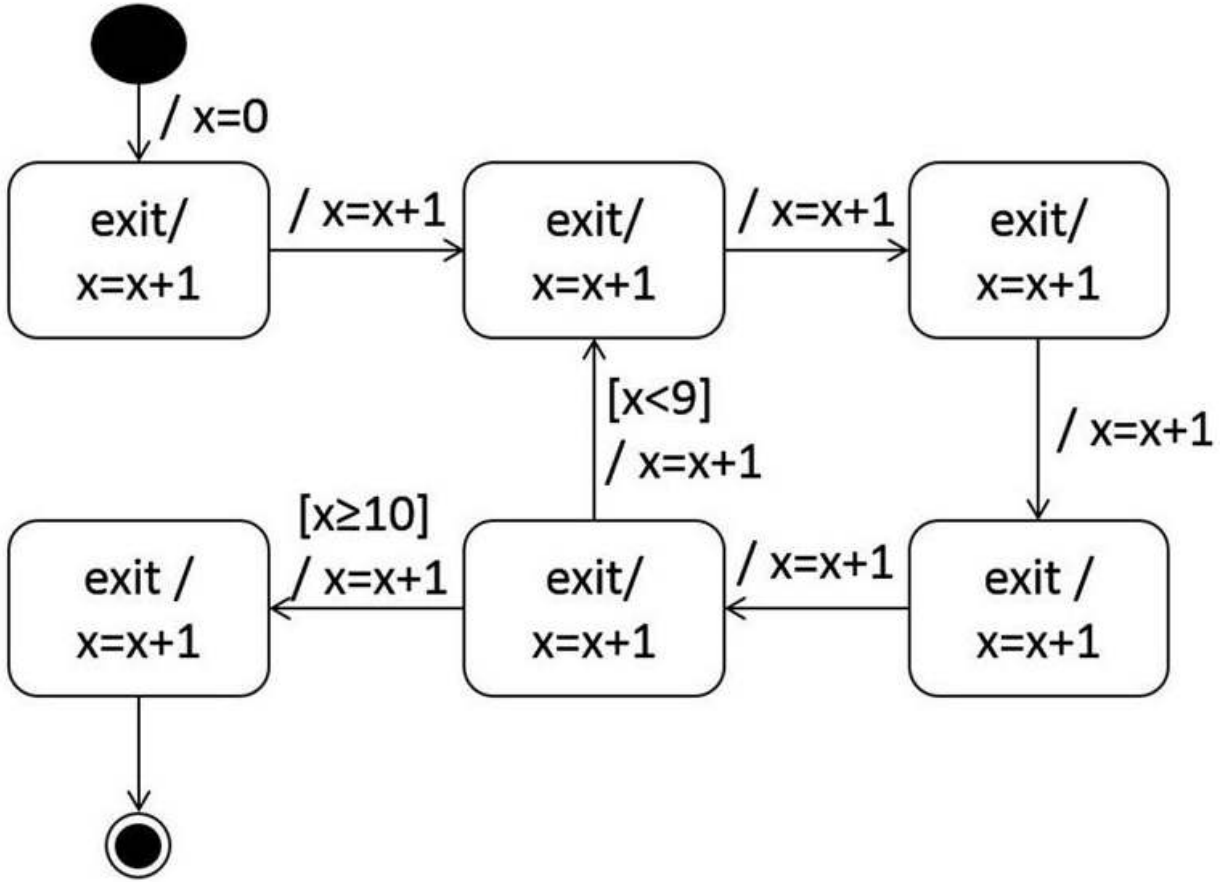
- A. It is not possible to tell.
- B. S1
- C. S2
- D. S3**
- E. The diagram is not a valid State Machine Diagram.

Discussion:

Possible Starting State	After event e1 next State	After Event e2 Next State	After Event e3 Next State	After Event e1 Next State
S1	S2	S2	S3	<u>S3</u>
S2	S2	S2	S3	<u>S3</u>
S3	S3	S3	S3	<u>S3</u>

No matter which state you start with, you wind up in S3.

Q43) After this state machine completes, what is the value of X?



- A. 11
B. 10
C. The state machine never ends.
D. 17
E. 20
F. 19

Discussion:

The [guard condition] is evaluated to determine if the transition should be taken. If the transition causes an exit to be taken, then the exit action is performed on the way out.

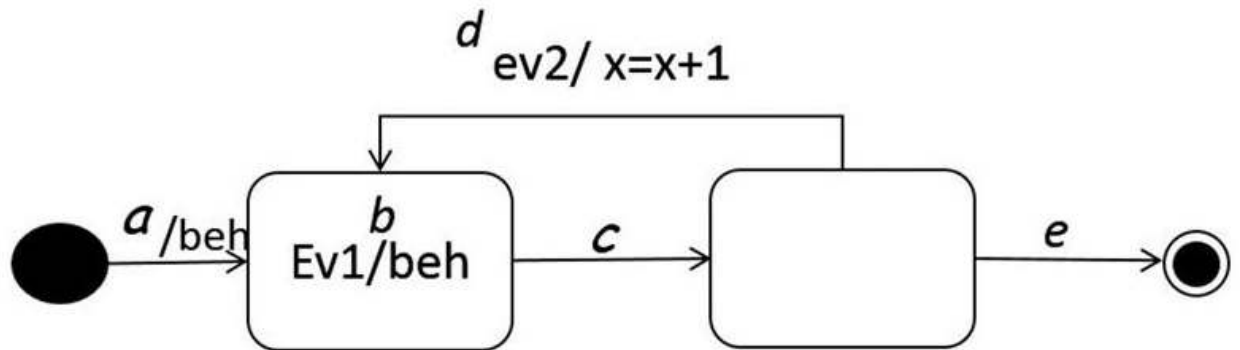
This is followed by the transition and the associated action.

In this type of question, a good strategy would be to count it twice. If you get the same answer, go with it. If not, come back to the question with a fresh mind.

F—The correct answer is 19.

In some situations, it may be considered that such values are undefined after the machine ends.

Q44) In the STM diagram fragment below, we see five transitions marked with italic letters. Which option cannot have a guard condition added?

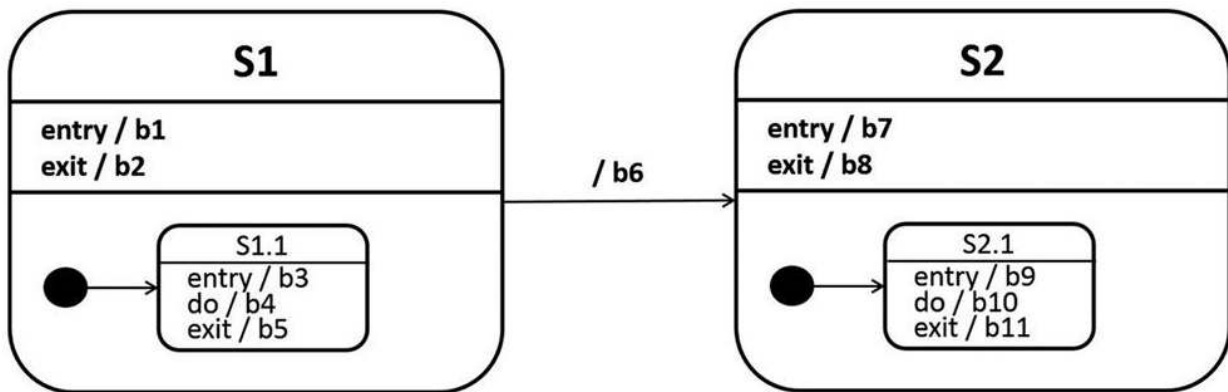


- A. *a*
- B. *b*
- C. *c*
- D. *d*
- E. *e*

Discussion:

- A—Correct. This is an initial transition. As a pseudostate, it is not allowed to have an event nor a guard because processing must not stop there. It can have an effect (behavior or action).
- B—Incorrect. This is an internal transition. It can have a trigger, guard, and effect.
- C—Incorrect. This is a completion (automatic) transition. It can have guard and effect but no trigger.
- D—Incorrect. This is a normally triggered transition. It can have a trigger, guard, and effect.
- E—Incorrect. This is a final transition. It can have a trigger, guard, and effect.

Q45) Assume the below state machine fragment is in State S1.1 when the do activity b4 finishes. What is the sequence of behaviors that then follows?



- A. b1, b2, b3, b4, b5, b6, b7, b8, b9, b10, b11
- B. The diagram is malformed.
- C. b5, b2, b7, b9
- D. b5, b2, b7, b9, b10
- E. b2, b5, b6, b7, b9, b10
- F. b5, b2, b6, b7, b9, b10**

Discussion:

- A—Incorrect. This is just the list of possible behaviors in collating order.
- B—Incorrect. The diagram is valid. For example, the transition between the states is a completion transition and needs no explicit trigger.
- C—Incorrect. The list of behaviors is missing b6. This is executed during the transition between S1.1 and S2.1. It is also missing the do activity behavior b10.
- D—Incorrect. The list of behaviors is still missing b6. This is executed during the transition between S1.1 and S2.1.
- E—Incorrect. The order of the behaviors is incorrect. When leaving states, the exit behaviors are done *inside out*. That is, the exit behavior from the lowest state that is exited is executed first, and the exit behavior from the highest state is executed last.
- F—Correct. Note that when entering states, the entry behaviors are done *outside in*. That is, the entry behavior from the highest state is executed first, and the entry behavior from the lowest state is executed last.

Q46) When discussing an Activity Diagram, what is a token?

- A. A token is used to get you on a subway in some cities.
- B. A token is a small symbol that looks like a ball with an arrowhead.
- C. A token represents an object or focus of control that traverses Activity Diagram edges or resides at a node.**
- D. A token represents a guard condition on an Activity Diagram.

Q47) An Activity has an Activity Final Node, with more than one incoming edge. When does the Activity finish?

A. The Activity finishes when all the incoming edges have supplied at least one token.

B. The Activity finishes when at least one incoming edge has supplied one token.

C. The Activity finishes when the majority of incoming edges have supplied at least one token.

D. The Activity finishes when at least one incoming edge has supplied one token and no token is moving in other parts of the diagram.

E. The Activity finishes when at least one incoming edge reaches a Flow Final Node, not an Activity Final Node.

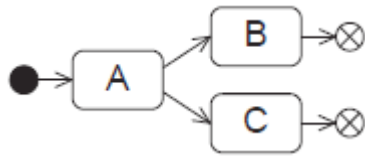
Discussion:

B—Yes, the first token that reaches an Activity Final Node terminates the diagram. This is different from most other Nodes in Activity Diagrams, which requires all incoming edges to supply tokens.

A, C and D—No, only one token from one edge is sufficient.

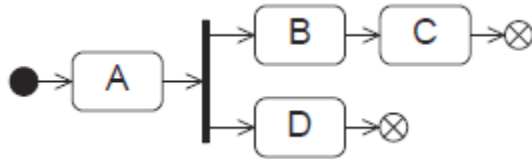
E—No, a Flow Final Node can terminate a diagram if there are no other active tokens in the diagram. Only an Activity Final Node will always terminate a diagram when a token arrives.

Q48) You are given the following activity diagram. Which of the following action sequences are possible during one execution of the activity diagram?



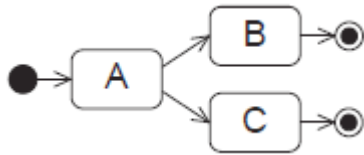
A	A → C <i>After the execution of C, the execution of this one branch is terminated, but the execution of the other branch can go on. Thus B has to be executed ensuring that the second flow final node is reached and the graph is terminated.</i>
<u>B</u>	A → C → B
C	A → B <i>After the execution of B, the execution of this one branch is terminated, but the execution of the other branch can go on. Thus C has to be executed ensuring that the second flow final node is reached and the graph is terminated.</i>
<u>D</u>	A → B → C

Q49) You are given the following activity diagram. Which of the following action sequences are possible during one execution of the activity diagram? (MULTIPLE ANSWER)



A	A → D
B	A → B → D
C	A → B → C → D
D	A → B → D → C
E	A → D → C
F	A → D → B → C
G	A → B → C

Q50) You are given the following activity diagram. Which of the following action sequences are possible during one execution of the activity diagram? (MULTIPLE ANSWER)



A	A → C → B
B	A → C
C	A → B
D	A → B → C